Finite adaptive secret-key converses for the amplitude-damping channel

Abstract

We prove strong, weak, binary-entropy, and correctness/secrecy converses for finite adaptive qubit amplitude-damping protocols with a charged independent classical seed. The final key fidelity is optimized against a state on the actual complete Eve register, containing every channel environment, the entire immutable public transcript, and the prior Eve register. Optimized normalized separable comparison flags give the strong exponent. An extensive privacy-test bound gives the weak and entropy inequalities. All four conclusions extend to the actual output closure at fixed budgets and fixed finite complete Eve.

The converse has rate threshold $C_Z(p) = \log_2[1 + (1-p)p^{p/(1-p)}]$, the capacity of the classical Z-channel. In particular $C_Z(1/2) = \log_2(5/4) \simeq 0.321928$. The argument builds on the private-state formulation of Horodecki, Horodecki, Horodecki, and Oppenheim [7], the privacy-test converse of Wilde, Tomamichel, and Berta [6], and the amortized framework of Kaur and Wilde [3]. The limit of the comparison rate as its order parameter tends to one is $\log_2(2-p)$, the known max-relative entanglement value of Rigovacca et al. [5, Proposition 2]. The improvement here comes from the order-dependent finite-code bounds and their $C_Z(p)$ limit at order parameter zero, with the seed and Eve specified in the model below.

1 The finite physical model and the four converses

All Hilbert spaces in the construction are finite dimensional over $\mathbb{C}$, and all logarithms are base two. A density operator is positive semidefinite and has trace one. Fix $0 < p < 1$, put $q = 1 - p$, and define

$$A_0 = |0\rangle\langle 0| + \sqrt{q}|1\rangle\langle 1|, \qquad A_1 = \sqrt{p}|0\rangle\langle 1|, \qquad \mathcal{A}_p(X) = \sum_{j=0}^{1} A_j X A_j^\dagger. \tag{1}$$

A physical call moves Alice's input qubit to Bob while leaving both old memories untouched. Its actual dilation is

$$V_p = \sum_{j=0}^{1} A_j \otimes |j\rangle_{E_{\mathrm{call}}}. \tag{2}$$

The environment in (2) is retained by Eve. It is a quantum register: off-diagonal terms between different physical loss labels are not dephased.

A protocol is a finite tree of complete local instruments, public outcome-dependent continuations, and physical calls. At a local Alice node the operators K_j satisfy $\sum_j K_j^\dagger K_j = I$; outcome j is sent publicly and selects the continuation. The Bob definition is analogous. Each private Kraus index can be retained coherently in the owning laboratory before a public outcome is sent. Local dimensions may change. The maximum call count is zero at a terminal node, the maximum of the continuation counts at a local node, and one plus the continuation count at a physical call. It is bounded by an integer $m \geq 0$ on *every* history. Complete instruments include all failure outcomes.

Eve's history space is defined recursively. It is a singleton at a terminal node, the direct sum $\bigoplus_j E(P_j)$ at a public local outcome, and $\mathbb{C}^2 \otimes E(P_1)$ at a physical call. A prior Eve register E_0 is included as an additional tensor factor. The actual honest–Eve output is obtained by summing all public branches with a classical copy of j in Eve's transcript, and by applying (2) at each call. Equivalently, the call recursion acts on the block $A_j X A_{j'}^\dagger$ for each pair of Eve labels (j, j'). Publicly

distinct histories have zero off-diagonal blocks. Physically distinct loss histories within the same public history retain their quantum coherences. This fixes the full Eve register rather than a selected marginal of it.

The initial state is exactly

$$\left(\sum_{j=1}^{D_0} a_j |jj\rangle\langle jj|_{J_A J_B}\right) \otimes \rho_{A_0} \otimes \rho_{B_0} \otimes \rho_{E_0}, \quad a_j \geq 0, \quad \sum_j a_j = 1, \quad D_0 \leq 2^k. \tag{3}$$

Here D_0 is a positive integer, $k \in \mathbb{R}$ is the alphabet budget, and the seed is independent of both local ancillas and prior Eve. There is no additional shared entangled input. Any local mixed ancilla or local private randomness can be purified within its own laboratory. The actual seed value is not announced as a public outcome.

At the terminal node, Alice and Bob choose basis-label functions with values in $\{1, \ldots, d\}$, measure those labels privately, and discard local work. The full output is a normalized classical–classical–quantum state

$$\rho_{K_A K_B E} = \sum_{i,j=1}^{d} |ij\rangle\langle ij| \otimes \rho_E^{ij}, \qquad \rho_E^{ij} \geq 0, \quad \sum_{i,j} \operatorname{Tr} \rho_E^{ij} = 1. \tag{4}$$

Arbitrary finite local processing can precede this readout. Every outcome, including a failure or unequal key pair, is included in (4). Define

$$\begin{aligned} C_d &= \frac{1}{d} \sum_{i=1}^{d} |ii\rangle\langle ii|, \\ F(\rho, \sigma) &= \|\sqrt{\rho}\sqrt{\sigma}\|_1^2, \qquad G(\rho) = \sup_{\xi_E \text{ density}} F(\rho, C_d \otimes \xi_E). \end{aligned} \tag{5}$$

The ideal comparison ξ_E is normalized on the *same complete* Eve space. The actual disagreement and Alice-key secrecy errors are

$$\epsilon(\rho) = \sum_{i \neq j} \operatorname{Tr} \rho_E^{ij}, \qquad \delta(\rho) = \frac{1}{2} \left\| \rho_{K_A E} - \frac{I_{K_A}}{d} \otimes \rho_E \right\|_1. \tag{6}$$

The marginal ρ_E in (6) is the actual one.

For $0 < z < 1$, set

$$B_z(p) = \log\left[1 + \left(\frac{q}{[1 - p^{1/(1-z)}]^{1-z}}\right)^{1/z}\right], \qquad C_Z(p) = \log[1 + qp^{p/q}]. \tag{7}$$

For $x \in [0, 1]$, write

$$L_d(x) = \sup_{\lambda \geq 0} \left\{ \lambda x - \log\left[1 + \frac{2^\lambda - 1}{d}\right] \right\}, \qquad h(x) = -x \log x - (1 - x) \log(1 - x). \tag{8}$$

The function h uses its continuous endpoint values.

Theorem 1 (Finite seed-key converses)**.** *Let $0 < p < 1$, $m \in \mathbb{N}_0$, $d \geq 2$, and $k \in \mathbb{R}$. For every finite code in the model above, let ρ be its complete-Eve output and let $G = G(\rho)$. Then for every*

$0 < z < 1$,

$$G \le \min\{1, 2^{-z(\log d - k - mB_z(p))}\}, \tag{9}$$

$$L_d(G) \le k + mC_Z(p), \tag{10}$$

$$G \log d - h(G) \le k + mC_Z(p), \tag{11}$$

$$\epsilon(\rho) + \delta(\rho) \ge 1 - \min\{1, 2^{-(z/2)(\log d - k - mB_z(p))}\}. \tag{12}$$

Consequently (12) *also holds with any larger explicit correctness and secrecy tolerances in place of the actual errors.*

Theorem 2 (Actual fixed-budget finite-Eve output closure)**.** *Fix the parameters of Theorem 1 and a finite Eve space E. Let $\mathcal{F}_{p,m,d,k}(E)$ consist of the actual code outputs placed in E by a* bijective *register relabeling of the complete space $E(P) \otimes E_0$. Define*

$$\mathcal{O}_{p,m,d,k}(E) = \{\rho : \ \forall \eta > 0 \ \exists \tau \in \mathcal{F}_{p,m,d,k}(E), \ \|\rho - \tau\|_1 < \eta\}. \tag{13}$$

All four inequalities of Theorem 1 hold for every density $\rho \in \mathcal{O}_{p,m,d,k}(E)$, with its own actual full-Eve marginals and errors. The parameters p, m, d, k, E are fixed in (13)*, and z is fixed when applying a strong inequality.*

For a sequence of these finite codes with $m \to \infty$, seed budgets $k_m = o(m)$, and correctness and secrecy errors tending to zero, $G_m \to 1$ by the fidelity–trace-distance inequalities. Equation (11) and $h(G_m) \le 1$ therefore give

$$\limsup_{m\to\infty} \frac{\log d_m}{m} \le C_Z(p).$$

The bound is thus a rate converse for the stated seed-assisted finite-code model, as well as a finite-error inequality.

Comparison with earlier bounds

Let h_2 denote binary entropy in bits. Two analytic benchmarks are the reverse-coherent-information achievable rate

$$L_{\mathrm{RCI}}(p) = \max_{0\le u\le 1} [h_2(u) - h_2(pu)]$$

and the balanced-squashing upper bound

$$U_{\mathrm{sq}}(p) = \max_{0\le u\le 1} [h_2((1-p/2)u) - h_2(pu/2)].$$

These appear in [8, Supplementary Note 5, Eqs. (S212), (S228)–(S231)]. The achievable entanglement rate is also an achievable secret-key rate. For comparison with the finite-code rate consequence proved above, at $p = 1/2$ the values per physical channel use are

Quantity	Value
Reverse coherent information L_{RCI}	$0.271553\ldots$
Present converse threshold $C_Z = \log_2(5/4)$	$0.321928\ldots$
Balanced-squashing bound $U_{\mathrm{sq}} = \log_2(4/3)$	$0.415037\ldots$
Max-relative bound $\log_2(3/2)$ [5]	$0.584963\ldots$

The first value is the one-variable maximum above. The third is exact: $h_2(3u/4) - h_2(u/4)$ is strictly concave on $(0,1)$, and its derivative vanishes at $u = 4/9$. The substitution $u = 1/2$ in [8, Eq. (S232)] instead gives $0.410869\ldots$; it does not maximize the preceding expression. Pirandola already identified this issue and evaluated the retained maximization [11, footnote 3 and Theorem IV.1]. Thus $C_Z(1/2)$ improves both analytic upper benchmarks in the table while leaving a gap above the achievable rate. Fawzi, Shayeghi, and Ta also provide a hierarchy of semidefinite strong-converse bounds for assisted quantum capacity [9, Sec. 6]; the table does not rank that hierarchy.

There is a particularly small asymptotic gap at high damping. Abdelhadi et al. give the triple-rail achievable rate $L_3(p) = (1-p)\log_2(3)/3$ [10, Sec. 2.2]. Writing $q = 1 - p \downarrow 0$, the elementary expansion $p^{p/q} = e^{-1}(1 + O(q))$ gives

$$C_Z(p) = \frac{q}{e \ln 2} + O(q^2), \qquad \lim_{p\uparrow 1}\left(\frac{C_Z(p)}{L_3(p)} - 1\right) = \frac{3}{e\ln 3} - 1 = 0.00457489\ldots.$$

The converse threshold is therefore about 0.4575% above this known achievable rate in the high-damping limit. This is a relative asymptotic comparison, not a uniform gap for every p.

2 Sandwiched quantities and supported comparison order

For $\alpha > 1$, write $t = (\alpha - 1)/\alpha$ and define

$$\begin{aligned} \widetilde{Q}_\alpha(X\|Y) &= \mathrm{Tr}[(Y^{-t/2}XY^{-t/2})^\alpha], \\ \widetilde{D}_\alpha(\rho\|\sigma) &= \frac{1}{\alpha-1}\log \widetilde{Q}_\alpha(\rho\|\sigma), \\ \widetilde{E}_\alpha(A:B)_\rho &= \inf_{\sigma\in \mathrm{SEP}(A:B)} \widetilde{D}_\alpha(\rho\|\sigma). \end{aligned} \tag{14}$$

Inverse powers act on the support. Unsupported comparisons have divergence $+\infty$. The separable set consists of normalized finite convex combinations of product densities across the entire honest cut. The sandwiched data-processing theorem for every $\alpha > 1$ is [1, 2]

$$\widetilde{Q}_\alpha(\Lambda(X)\|\Lambda(Y)) \le \widetilde{Q}_\alpha(X\|Y) \tag{15}$$

for CPTP Λ and supported positive X, Y. Normalizing each positive input separately establishes the unnormalized version, using trace preservation and $\widetilde{Q}_\alpha(aX\|bY) = a^\alpha b^{1-\alpha}\widetilde{Q}_\alpha(X\|Y)$.

For positive X, Y, Z, if $Y \le cZ$ with $c > 0$ and $\mathrm{supp}\, X \subseteq \mathrm{supp}\, Y$, then

$$\widetilde{Q}_\alpha(X\|Z) \le c^{\alpha-1}\widetilde{Q}_\alpha(X\|Y). \tag{16}$$

Indeed Löwner–Heinz gives $Y^t \le c^t Z^t$ and makes $C = c^{-t/2}Z^{-t/2}Y^{t/2}$ a contraction. Support containment implies $Z^{-t/2}XZ^{-t/2} = c^t C(Y^{-t/2}XY^{-t/2})C^\dagger$. The Schatten-$\alpha$ ideal inequality proves (16). This uses no commutativity. Mixing any supported separable σ with a full-rank product density ζ gives

$$\widetilde{D}_\alpha(\rho\|(1-\eta)\sigma + \eta\zeta) \le \widetilde{D}_\alpha(\rho\|\sigma) - \log(1-\eta), \quad 0 < \eta < 1. \tag{17}$$

Thus full-rank comparisons on the original input space approximate the infimum. A finite scalar domination $\rho \le C_0\sigma$ then propagates all input support conditions through CP branches.

For later use, recording a complete instrument (R_j) and then discarding a direct-sum flag gives

$$\sum_j \widetilde{Q}_\alpha(R_j X R_j^\dagger \| R_j Y R_j^\dagger) \le \widetilde{Q}_\alpha(X\|Y),$$

$$\widetilde{Q}_\alpha\left(\sum_j X_j \Big\| \sum_j Y_j\right) \le \sum_j \widetilde{Q}_\alpha(X_j\|Y_j). \tag{18}$$

Both inequalities follow from (15) and the exact direct-sum formula. Zero branches are retained with zero contribution.

3 Finite loss sectors and the phase channel

Fix $n \ge 1$. Identify computational strings with subsets of $[n]$ and let $H_w = \mathrm{span}\{|S\rangle : |S| = w\}$. For $0 \le r \le w \le n$, put

$$K_L|S\rangle = \begin{cases} \binom{w}{r}^{-1/2} |S \setminus L\rangle, & L \subseteq S,\ |L| = w-r, \\ 0, & \text{otherwise}, \end{cases} \qquad \mathcal{N}_{w,r}(X) = \sum_{|L|=w-r} K_L X K_L^\dagger. \tag{19}$$

Each input S occurs for $\binom{w}{r}$ loss sets. For a fixed L, distinct inputs containing L have distinct survivors. It follows that $\sum_L K_L^\dagger K_L = I_{H_w}$ and

$$\mathcal{A}_p^{\otimes n}\big|_{H_w} = \bigoplus_{r=0}^{w} \pi_{w,r} \mathcal{N}_{w,r}, \qquad \pi_{w,r} = \binom{w}{r} q^r p^{w-r}. \tag{20}$$

The survivor spaces are orthogonal, and the probabilities are input-independent and sum to one. This is a formula for the honest channel after tracing its quantum loss environment; Eve's physical dilation remains (2) throughout the actual code.

Let $\omega = e^{2\pi i/3}$ and choose $a \in \{0,1,2\}^n$ uniformly. Set

$$|\chi_a\rangle = \sum_{|S|=w} \omega^{-\sum_{i\in S} a_i} |S\rangle, \qquad |b_a\rangle = \binom{n}{r}^{-1/2} \sum_{|T|=r} \omega^{-\sum_{i\in T} a_i} |T\rangle,$$

and define the finite measure-and-prepare channel

$$\mathcal{S}_{w,r}(X) = 3^{-n} \sum_a \langle\chi_a|X|\chi_a\rangle |b_a\rangle\langle b_a|. \tag{21}$$

Character orthogonality gives $3^{-n}\sum_a |\chi_a\rangle\langle\chi_a| = I_{H_w}$, and b_a is normalized. Thus this is a complete CPTP channel. Measuring at Alice, sending the finite classical outcome, and preparing b_a at Bob implements it with only local operations and classical communication. It preserves the separable cone on every whole-memory cut.

Its unnormalized Choi matrix separates into positive rank-one blocks labeled by $1_S - 1_T$ modulo three. The coordinates $-1, 0, 1$ are distinct modulo three. Signatures without negative coordinates occur exactly when $T \subseteq S$, with $L = S \setminus T$; their coefficients are $1/\binom{n}{r}$, whereas the physical normalized loss coefficients are $1/\binom{w}{r}$. The remaining signatures give disjoint positive blocks. Consequently

$$\mathcal{N}_{w,r} \le_{\mathrm{CP}} c_{n,w,r} \mathcal{S}_{w,r}, \qquad c_{n,w,r} = \frac{\binom{n}{r}}{\binom{w}{r}} \ge 1. \tag{22}$$

Choi positivity proves complete positivity of the remainder, so (22) remains true with arbitrary memory identities. Combining (22), (16), and data processing gives

$$\widetilde{Q}_\alpha(\mathcal{N}_{w,r}(X)\|\mathcal{S}_{w,r}(Y)) \le c_{n,w,r}^{\alpha-1}\widetilde{Q}_\alpha(X\|Y) \tag{23}$$

on supported positive inputs including all side memories.

4 Optimized normalized flags and coherent input records

Fix $0 < z < 1$ and set

$$\alpha = \frac{1}{1-z} > 1, \qquad \alpha - 1 = \alpha z. \tag{24}$$

The following comparison flags are functions only of n, w, p, z:

$$Z_w(z) = \sum_{r=0}^{w} \pi_{w,r} c_{n,w,r}^z, \qquad \mu_{w,r} = \frac{\pi_{w,r} c_{n,w,r}^z}{Z_w(z)}. \tag{25}$$

They are positive and sum to one; $Z_w(z) \ge 1$. The physical flag law continues to be $\pi_{w,r}$. For these *comparison* masses,

$$\sum_r \pi_{w,r}^\alpha \mu_{w,r}^{1-\alpha} c_{n,w,r}^{\alpha-1} = Z_w(z)^\alpha. \tag{26}$$

To verify it, substitute (25): the power of c becomes $(\alpha-1)(1-z) = z$, and the power of π becomes one. More generally Hölder gives $\sum_r a_r \mu_r^{1-\alpha} \ge (\sum_r a_r^{1/\alpha})^\alpha$ for $a_r = \pi_r^\alpha c_r^{\alpha-1}$, with equality at $\mu_r \propto a_r^{1/\alpha} = \pi_r c_r^z$. Only this explicit normalized comparison is needed below.

For an arbitrary whole-cut input, Alice makes the local basis record $V|S\rangle = |S\rangle|S\rangle_W$ before the block channel. Write Ω_X for the recorded physical output. Fourier measurement of W in the binary basis, public announcement of its label b, and Bob's Z^b phase correction define a complete recovery map $\mathcal{R}$ with

$$\mathcal{R}(\Omega_X) = (\mathrm{id} \otimes \mathcal{A}_p^{\otimes n})(X). \tag{27}$$

For the matrix unit $|S\rangle\langle T|$ and a common lost set L, the measured branch has factor $2^{-n}(-1)^{b\cdot(S-T)}$; the surviving strings $S \setminus L, T \setminus L$ have the same phase difference. Bob's correction cancels it and the sum of all 2^n outcomes is exactly the original output. Operator-valued coefficients on old memories cause no change to this calculation.

Pinch W by its $n+1$ weight projections and call this map $\mathcal{P}$. Let X_w, Y_w be the recorded weight branches of supported positive inputs X, Y. Their channel inputs are in H_w. Define

$$\kappa_Y = \bigoplus_{w,r} \mu_{w,r} \mathcal{S}_{w,r}(Y_w), \qquad \mathcal{T}_{n,p,z}(Y) = \mathcal{R}(\kappa_Y). \tag{28}$$

The w record is Alice's; survivor flags are Bob's. These flags may be retained explicitly or represented by their orthogonal subspaces, then discarded by complete local maps. All constituent maps in (28) are finite and complete. Thus $\mathcal{T}_{n,p,z}$ is a literal CPTP separability-preserving channel on the block's original laboratory coordinates.

The direct-sum formula, (23), and (26) give

$$\begin{aligned} \widetilde{Q}_\alpha(\mathcal{P}\Omega_X\|\kappa_Y) &\le \sum_w \widetilde{Q}_\alpha(X_w\|Y_w) Z_w(z)^\alpha \\ &\le \big(\max_w Z_w(z)\big)^\alpha \widetilde{Q}_\alpha(X\|Y). \end{aligned} \tag{29}$$

The last step is the complete recorded input instrument in (18). In particular, actual and comparison input-weight probabilities are charged by this same direct-sum formula rather than by selective averages of an entanglement measure.

The pinching order $\Omega_X \le (n+1)\mathcal{P}\Omega_X$ gives support containment in κ_Y. Congruence by $\kappa_Y^{-z/2}$ and positive eigenvalue monotonicity yield $\widetilde{Q}_\alpha(\Omega_X\|\kappa_Y) \le (n+1)^\alpha \widetilde{Q}_\alpha(\mathcal{P}\Omega_X\|\kappa_Y)$. Use recovery data processing to obtain

$$\widetilde{Q}_\alpha((\mathrm{id}\otimes\mathcal{A}_p^{\otimes n})(X)\|\mathcal{T}_{n,p,z}(Y)) \le F_{n,z}\widetilde{Q}_\alpha(X\|Y), \quad F_{n,z} = (n+1)^\alpha(\max_w Z_w(z))^\alpha. \tag{30}$$

The bit charge is consequently

$$C_{n,z}(p) = \frac{\log F_{n,z}}{\alpha-1} = \frac{1}{z}\big[\log(n+1) + \log\max_w Z_w(z)\big]. \tag{31}$$

All statements hold on arbitrary finite side-memory spaces and all supported positive subnormalized branches. The recovery and comparison operations concern honest states used in the bound; they never replace the actual channel environment in the key criterion.

5 The scalar estimate and the order limit

For $w > 0$, put $u = w/n$, $v = r/w$, and let $D_{\mathrm{bin}}(v\|q) = v\log(v/q) + (1-v)\log((1-v)/p)$. The binomial bound $\binom{N}{k} \le 2^{Nh(k/N)}$ follows by bounding a binomial probability at parameter k/N by one. Therefore

$$\begin{aligned}\pi_{w,r}c_{n,w,r}^z &= \binom{n}{r}^z\binom{w}{r}^{1-z} q^r p^{w-r} \\ &\le 2^{nz[\,,h(uv)-uh(v)-(u/z)D_{\mathrm{bin}}(v\|q)\,]}.\end{aligned} \tag{32}$$

Both binomial exponents are nonnegative. The $w = 0$ sum is one. There are at most $n+1$ terms, so

$$C_{n,z}(p) \le nB_z(p) + \frac{2}{z}\log(n+1), \tag{33}$$

once the maximum of the expression in brackets is evaluated.

For $v > 0$, set $J(v) = [h(v) + D_{\mathrm{bin}}(v\|q)/z]/v$ and $x = uv$. The objective is $h(x) - xJ(v)$ on $0 \le x \le v$. Its derivative in u is positive infinite at zero and equals $\log(1-v) - D_{\mathrm{bin}}(v\|q)/z < 0$ at one. Thus the unconstrained entropy-conjugate maximum is admissible and is $\log(1 + 2^{-J(v)})$. Direct differentiation gives

$$J'(v) = \frac{\log p + (z-1)\log(1-v)}{zv^2}.$$

Its unique minimum is $v_z = 1 - p^{1/(1-z)}$, and $J(v_z) = ((1-z)\log v_z - \log q)/z$. Substitution gives exactly (7). The $v = 0$ boundary contributes at most zero. This proves (33) analytically.

For the weak bound the needed order limit is

$$\lim_{z\downarrow 0} B_z(p) = C_Z(p). \tag{34}$$

Indeed let $f(z) = \ln q - (1-z)\ln[1 - p^{1/(1-z)}]$. It satisfies $f(0) = 0$ and $f'(0) = \ln q + (p/q)\ln p$. The inner power in (7) is $\exp(f(z)/z)$, which tends to $qp^{p/q}$. This proves (34). The block overhead in (33) is removed at each fixed $z > 0$ before this order limit is taken.

6 Actual adaptive iteration and the charged preparation

For a finite n-qubit block tree R, evaluate a comparison tree using exactly the same local instruments and continuations, with each physical block replaced by $\mathcal{T}_{n,p,z}$. Call its output $R^{\mathrm{cmp}}[Y]$. It is a normalized separable state on the output cut whenever Y is normalized and separable. Support is preserved at each CP branch and block, and upon recombination.

Induction on the complete tree, using (18) at every local node and (30) at every block-call node, gives

$$\widetilde{Q}_\alpha(R[X]\|R^{\mathrm{cmp}}[Y]) \le F_{n,z}^{c(R)}\widetilde{Q}_\alpha(X\|Y). \tag{35}$$

At a local node first sum the continuation bounds, then replace their call exponents by the maximum using $F_{n,z} \ge 1$, and finally apply the complete input instrument bound. At a call node multiply the continuation factor by one additional $F_{n,z}$. Thus the induction uses the maximum call budget on every public history and all unnormalized branch masses. Approximating the input separable infimum using (17) and taking logarithms yields

$$\widetilde{E}_\alpha(R[\sigma]) \le e + mC_{n,z}(p) \quad \text{if } \widetilde{E}_\alpha(\sigma) \le e, \quad c(R) \le m. \tag{36}$$

We now construct the coherent upper preparation for (3). Use the vector

$$|a\rangle_{J_A J_B} = \sum_j \sqrt{a_j}|j\rangle|j\rangle. \tag{37}$$

Alice privately copies j into a local work register. Tracing that work recovers the classical seed of (3) exactly, while retaining it keeps a pure upper preparation. Bob and Alice retain local purifications of their independent ancillas. An independent prior Eve density may be purified with its additional purifier placed in Alice's work; tracing that purifier recovers precisely the given prior Eve. All subsequent original operations act as identities on these new work registers. Tracing them at the end gives the original honest–Eve output, entry by entry, including all its physical-loss coherences. No seed label is added to the public transcript.

The explicit normalized separable comparison for (37) is $\tau_0 = D_0^{-1}\sum_j |jj\rangle\langle jj|$. On the support of the seed vector it acts as I/D_0. The sandwiched operator is $D_0^t|a\rangle\langle a|$, and therefore

$$\widetilde{D}_\alpha(|a\rangle\langle a|\|\tau_0) = \log D_0 \le k. \tag{38}$$

The private copy is an Alice isometry; independent local factors do not increase this exhibited comparison cost. Tracing prior Eve is local to the Alice–Eve side for this input calculation and leaves the comparison separable on the honest cut. Hence the actual retained initial honest state has $\widetilde{E}_\alpha \le k$. For n independent preparations the tensor comparison gives an initial bound nk. This charges the actual seed alphabet without an entropy assumption on its law.

7 A coherent lift of the complete-Eve key output

Retain all local work and local copies of public records. At a public Alice node the global dilation is the sum over j of the original local operator times $|j\rangle_{H_A}|j\rangle_{H_B}|j\rangle_E$; analogously for Bob. Tracing Eve dephases the honest copies, so this honest retained operation is a complete local instrument followed by an ordinary public message. The copies do not supply coherent nonlocal communication. Channel loss labels are retained exactly through (2).

Coherently implement the terminal local labels and retain orthogonal local key copies C_A, C_B. A purification of the complete output can be written

$$|\psi\rangle = \sum_{i,j} |ij\rangle_{K_A K_B} |ij\rangle_{C_A C_B} |\chi_{ij}\rangle_{S'E}. \tag{39}$$

The vectors are subnormalized and $\rho_E^{ij} = \mathrm{Tr}_{S'} |\chi_{ij}\rangle\langle\chi_{ij}|$. Tracing honest work gives the original (4); tracing Eve gives the retained coherent honest state $\widehat{\rho}$. Public record copies and private key copies are part of the honest shields S_A, S_B. The lift is the usual finite private-state construction, with squared fidelity as in [3, Sec. 3.3].

Fix a normalized ξ_E. Direct-sum fidelity gives

$$f(\rho, C_d \otimes \xi_E) = \frac{1}{\sqrt{d}} \sum_i \| \sqrt{\rho_E^{ii}} \sqrt{\xi_E} \|_1, \qquad f = \sqrt{F}. \tag{40}$$

After a finite local enlargement of the shields, Uhlmann's theorem chooses for each i a normalized purification $|\eta_i\rangle$ of the *same* ξ_E, in the key-copy subspace labeled ii, whose overlap with the actual branch is the corresponding nonnegative summand in (40). Fix a purification $|\theta\rangle_{SE}$ of ξ_E. Extending the purifying isometries to finite shield unitaries gives $|\eta_i\rangle = U_i|\theta\rangle$. Define a controlled twisting unitary $U = \sum_{i,j} |ij\rangle\langle ij| \otimes U_{ij}$ with $U_{ii} = U_i$ and arbitrary unitary choices on the other blocks. Then

$$|\Gamma\rangle = \frac{1}{\sqrt{d}} \sum_i |ii\rangle U_i |\theta\rangle, \qquad \gamma = \mathrm{Tr}_E |\Gamma\rangle\langle\Gamma| = U(\Phi_d \otimes \theta_S) U^\dagger \tag{41}$$

is a private state. Its overlap with (39) is exactly (40), so fidelity monotonicity under tracing Eve gives

$$F(\widehat{\rho}, \gamma) \geq F(\rho, C_d \otimes \xi_E). \tag{42}$$

The twisting unitaries define an ideal mathematical target and are not additional operations performed by the original protocol. The shield enlargement is finite because ξ_E has finite rank. Taking candidates approaching the supremum in (5) is sufficient throughout.

The privacy-test projection for this target is $\Pi = U(\Phi_d \otimes I_S) U^\dagger$. It passes γ with probability one. Measuring this projection in (42) gives

$$\mathrm{Tr}\, \Pi \widehat{\rho} \geq F(\rho, C_d \otimes \xi_E). \tag{43}$$

For a normalized product vector with local expansions $|a\rangle = \sum_i |i\rangle|a_i\rangle$ and $|b\rangle = \sum_j |j\rangle|b_j\rangle$, its acceptance is

$$\frac{1}{d} \left\| \sum_i U_{ii}^\dagger (|a_i\rangle \otimes |b_i\rangle) \right\|^2 \leq \frac{1}{d} \left(\sum_i \|a_i\| \|b_i\| \right)^2 \leq \frac{1}{d}. \tag{44}$$

The first inequality is the triangle inequality and the second is Cauchy–Schwarz. Convexity extends (44) to every normalized separable state, with no factor depending on the shield dimensions. It also applies to tensor products of privacy tests as a private test of the product key dimension.

8 Finite independent lanes and the strong bound

Fix one original code and prepare n independent copies, each with its own seed and local preparations. Pad every history to m physical calls with a fresh ground-state input. A padding call has output ground state and ground-state environment, because $V_p|0\rangle = |0\rangle|0\rangle$. Its local preparation and

discard have only a constant public outcome. The expanded public history is determined by the original history; adding its padding labels and the zero loss labels embeds the original full Eve space isometrically into the padded space.

Optimal ideal-key fidelity is unchanged by this Eve isometry. For an arbitrary candidate on the larger space, compress it to the isometry's image and normalize. If its compressed trace is $r \le 1$, its root fidelity to the supported actual state equals $\sqrt{r}$ times the normalized candidate fidelity. Normalizing therefore cannot decrease that candidate fidelity. The reverse inequality follows by embedding an original candidate. This proves equality without discarding any actual Eve information.

Schedule the padded copies by scalar call number: execute each lane's original local segment until its next call, then apply one $\mathcal{A}_p^{\otimes n}$ block to their n inputs. All local feedback depends on the same original public outcomes. The resulting tree is finite and has at most m block calls. Retain both honest public copies. Local removal of unused padding ports and regrouping of the retained records recover $\widehat{\rho}^{\otimes n}$ from its honest output. All these changes are local coordinate isometries or complete local discards. A basis injection is decoded by its inverse on its image; on the orthogonal complement a fixed local output completes the decoder to a trace-preserving channel. The complement has zero probability on the constructed state. Thus the decoding does not introduce a resource premise or postselection.

Applying (36) with the independently charged input (38), and then local decoding data processing, gives

$$\widetilde{E}_\alpha(\widehat{\rho}^{\otimes n}) \le nk + mC_{n,z}(p). \tag{45}$$

The actual decoded key–Eve outputs are the independent product of the original full outputs, up to the described Eve isometry. For any fixed candidate ξ_E, let $g = F(\rho, C_d \otimes \xi_E)$ and use its private test from (43). The product test on the n retained outputs has acceptance at least g^n. Every separable comparison across the total two-laboratory cut has acceptance at most d^{-n} by (44). For $g > 0$, binary sandwiched data processing gives

$$\widetilde{E}_\alpha(\widehat{\rho}^{\otimes n}) \ge n \log d + \frac{n}{z} \log g. \tag{46}$$

The positive failure term in the binary $\widetilde{Q}_\alpha$ was omitted. A zero comparison acceptance gives infinite divergence and satisfies the same bound. The zero-fidelity case is immediate.

Take the supremum of candidates, combine (45)–(46), and use (33). For $G > 0$ the result is, for every finite $n \ge 1$,

$$\log d + \frac{1}{z} \log G \le k + mB_z(p) + \frac{2m}{z} \frac{\log(n+1)}{n}. \tag{47}$$

Let n increase with the original code and z fixed. The remainder vanishes. Exponentiation and $G \le 1$ prove (9), including $m = 0$. Every construction before this numerical limit is finite.

9 The extensive weak privacy-test bound

For the same private target, let $a = \operatorname{Tr} \Pi \widehat{\rho}$, so $a \ge g$. On n independent honest outputs measure the individual commuting projections $\Pi_1, \dots, \Pi_n$, and let T count successes. For any normalized separable comparison τ across the *total* honest cut and any subset $J \subseteq [n]$,

$$\operatorname{Tr} \tau \prod_{i \in J} \Pi_i \le d^{-|J|}. \tag{48}$$

The product is a private test of key dimension $d^{|J|}$; tracing the remaining lanes preserves separability. For $\lambda \geq 0$ the nonnegative expansion of the exponential moment then gives

$$\operatorname{Tr} \tau \, 2^{\lambda T} = \sum_{J \subseteq [n]} (2^{\lambda} - 1)^{|J|} \operatorname{Tr} \tau \prod_{i \in J} \Pi_i \leq \left[1 + \frac{2^{\lambda} - 1}{d} \right]^n . \tag{49}$$

No conditioning on a successful nonlocal test is involved.

Let P, Q be the two measured classical distributions. Their ordinary relative entropy obeys

$$D(P\|Q) \geq \lambda \, \mathbb{E}_P T - \log \mathbb{E}_Q 2^{\lambda T}. \tag{50}$$

For example, normalize $Q_x 2^{\lambda T(x)}$ to a probability law Q'; nonnegativity of $D(P\|Q')$ gives the identity and inequality in (50). If Q vanishes on positive P mass, the inequality holds with infinite divergence. For supported classical distributions and $\alpha > 1$, Jensen's inequality also gives

$$D(P\|Q) = \sum_x P_x \log(P_x/Q_x) \leq \frac{1}{\alpha - 1} \log \sum_x P_x (P_x/Q_x)^{\alpha - 1} = \widetilde{D}_{\alpha}(P\|Q). \tag{51}$$

The mean success count on the actual product output is $na \geq ng$. Thus sandwiched data processing, (49), and (50)–(51) prove

$$\widetilde{D}_{\alpha}(\widehat{\rho}^{\otimes n}\|\tau) \geq n \left\{ \lambda g - \log \left[1 + \frac{2^{\lambda} - 1}{d} \right] \right\}.$$

Infimize over the same normalized separable comparisons, take candidates approaching G, and then take the supremum in λ. This yields the extensive lower bound

$$n L_d(G) \leq \widetilde{E}_{\alpha}(\widehat{\rho}^{\otimes n}). \tag{52}$$

It is a direct ordinary-divergence test estimate bounded above by the sandwiched resource through (51).

Combine (52), (45), and (33). At each fixed $z > 0$, let n increase to obtain $L_d(G) \leq k + m B_z(p)$. Only after this finite-lane remainder has vanished, let z decrease to zero and apply (34). This proves (10).

For completeness the scalar conjugate in (8) is

$$L_d(x) = \begin{cases} 0, & 0 \leq x \leq 1/d, \\ x \log(dx) + (1 - x) \log \dfrac{1 - x}{1 - 1/d}, & 1/d < x \leq 1. \end{cases} \tag{53}$$

Differentiating the objective gives $x - 2^{\lambda}/(d - 1 + 2^{\lambda})$. If $x \leq 1/d$, its maximum is at zero. If $1/d < x < 1$, the maximum satisfies $2^{\lambda} = x(d-1)/(1-x)$; substitution gives (53). The case $x = 1$ follows as $\lambda \to \infty$. For $x > 1/d$, expanding the second clause gives $L_d(x) = x \log d - h(x) - (1 - x) \log(1 - 1/d) \geq x \log d - h(x)$. For $0 < x \leq 1/d$, $h(x) \geq x \log(1/x) \geq x \log d$; at zero it is immediate. Thus

$$L_d(x) \geq x \log d - h(x), \qquad 0 \leq x \leq 1. \tag{54}$$

Equations (10) and (54) prove (11).

10 Actual correctness and secrecy against whole Eve

For (4), correct Bob's label to Alice's mathematically by

$$\rho^{\mathrm{corr}} = \sum_i |ii\rangle\langle ii| \otimes \sum_j \rho_E^{ij}.$$

The wrong-label blocks contribute total trace ϵ; the added diagonal blocks contribute the same trace. Both differences are positive on their respective orthogonal key blocks, so

$$\tfrac{1}{2}\|\rho - \rho^{\mathrm{corr}}\|_1 = \epsilon(\rho). \tag{55}$$

The isometry $|i\rangle \mapsto |ii\rangle$ identifies ρ^{corr} with the actual Alice–Eve marginal. It also sends the uniform Alice state tensored with actual ρ_E to $C_d \otimes \rho_E$. Consequently

$$\tfrac{1}{2}\|\rho^{\mathrm{corr}} - C_d \otimes \rho_E\|_1 = \delta(\rho), \qquad \tfrac{1}{2}\|\rho - C_d \otimes \rho_E\|_1 \le \epsilon + \delta. \tag{56}$$

Fuchs–van de Graaf [4] gives $\sqrt{F(\rho, C_d \otimes \rho_E)} \ge (1 - \epsilon - \delta)_+$. The actual ρ_E is one of the candidates in (5), so

$$\epsilon + \delta \ge 1 - \sqrt{G}. \tag{57}$$

Taking the square root of (9) gives $\sqrt{G} \le \min\{1, 2^{-(z/2)(\log d - k - mB_z(p))}\}$. Combining this with (57) proves (12). The factor $1/2$ in the exponent follows from the squared fidelity convention; the Eve marginal in the secrecy error was never optimized or reduced. This completes Theorem 1.

11 Continuity on the actual fixed-Eve output family

Bijective relabeling of a complete Eve register preserves trace norm, fidelity, the actual marginal, and both errors. The ideal candidates are in bijection under the same relabeling. Thus every member of $\mathcal{F}_{p,m,d,k}(E)$ satisfies Theorem 1.

For densities ρ, τ on the same complete output space, let $u = \|\rho - \tau\|_1$. For every fixed density σ,

$$|F(\rho, \sigma) - F(\tau, \sigma)| \le 2\sqrt{u}, \qquad |G(\rho) - G(\tau)| \le 2\sqrt{u}. \tag{58}$$

To prove the first bound, Uhlmann and Fuchs–van de Graaf give purifications $|r\rangle, |t\rangle$ with $\|r - t\|^2 = 2(1 - f(\rho, \tau)) \le u$. For a purification of σ optimal against r, the difference of overlaps is at most $\|r - t\| \le \sqrt{u}$; the other overlap is bounded by $f(\tau, \sigma)$. Interchanging the states gives $|f(\rho, \sigma) - f(\tau, \sigma)| \le \sqrt{u}$. Since the two root fidelities are in $[0, 1]$, squaring gives the first inequality in (58). It is uniform in σ, so taking the two suprema over $C_d \otimes \xi_E$ gives the second inequality.

Every finite output is ccq. If $\tau_j \to \rho$ in trace norm, each off-diagonal key matrix entry of ρ is zero because its absolute value is bounded by $\|\rho - \tau_j\|_\infty \le \|\rho - \tau_j\|_1$. Thus ρ is also ccq. Its G is the limit of the finite-output values by (58). The functions L_d and h are continuous on $[0, 1]$ by (53). Hence (9), (10), and (11) pass to every output in (13). Finally (57) applies to this ccq limit with its actual full-Eve marginal, proving its (12). This proves Theorem 2.

The closure is explicitly the norm closure of actual finite outputs with fixed complete finite Eve and fixed numerical budgets. No continuity of entanglement measures in a growing shield dimension, and no membership claim for an additional infinite protocol model, is required.

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